

Executive Summary: Predicting Graft Acceptance/Rejection After Liver Transplantation

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Motivation and Objective:

Liver transplantation is the treatment for liver diseases with irreversible damage. Graft rejection remains a significant obstacle to long-term success. Early detection of graft rejection improves patient outcomes. Traditional diagnostic methods, such as Banff histology, are primarily based on histopathological evaluation. They are constrained by subjectivity, sampling bias, and limited capacity to capture the molecular complexity of immune-mediated injury. Recent advances in transcriptomic profiling have redefined rejection. The INTERLIVER study (“Defining an NK Cell–enriched Rejection-like Phenotype in Liver Transplant Biopsies”)¹ identified distinct molecular archetypes of liver allograft inflammation, including NK cell–mediated and T cell–mediated rejection signatures. These findings are highlighting the potential of gene expression analysis to provide a more objective and mechanistic assessment of graft status.

Building on these findings, our project aims to develop a computational model capable of predicting rejection status from gene expression data, using the INTERLIVER dataset as a reference. NK cell– and T cell–mediated phenotypes were labeled as rejection, while other archetypes were classified as non-rejection, providing a molecularly informed framework for model training and validation. Our objective is to establish a predictive approach that can reliably distinguish rejection from stable graft states, improving accuracy and reproducibility.

Stakeholders:

Predictive graft rejection model is appealing to the following parties:

- **Transplant Physicians / Hepatologists:** Use diagnostic tools to guide treatment decisions and immunosuppressive therapy.
- **Hospital and Transplant Centers:** Benefit from improved graft survival rates and reduced readmissions or complications.
- **Regulatory Bodies / Health Authorities:** Ensure any clinical diagnostic tool meets safety, accuracy, and compliance standards.

Data Sources and Exploration:

The CEL files are available on the Gene Expression Omnibus (GEO) website (GSE277334).

The full model classified 763 biopsies into the following molecular phenotypes, characterized by their gene expression:

- no-rejection 54%,
- TCMR 16%, (T cell–mediated rejection)
- NKRL 13%, (NK cell–enriched rejection-like)
- injury 16%.

Preprocessing:

Affymetrix microarray expression data was imported from Gene Expression Omnibus (GEO) for analyzing clinical biopsies. Data was imported with GEOparse and pandas, annotated with gene symbols, and cleaned by removing unannotated probes and collapsing duplicates using median expression values. Quality control included principal component analysis, Relative Log Expression assessment, and intensity-based filtering to ensure only reliable transcripts above the detection threshold are retained for downstream data modeling and statistical analyses

EDA:

- *Top Differentially Expressed Genes (t-test Analysis)*

A t-test was performed to identify genes that were significantly differentially expressed between rejection and non-rejection liver transplant biopsies. Adjusted p-values (FDR correction) were used to control for false discoveries due to multiple testing. The top five genes identified were GZMA, ADCY7, LCP2, FYB, and KLRK1.

All five genes are associated with immune activation and cytotoxic activity, which aligns with the biological processes underlying transplant rejection. This supports that our dataset and analysis capture meaningful immune-related transcriptional changes distinguishing rejection from non-rejection samples.

- *Principal Component Analysis (PCA)*

Before feature selection with t-test, PC1 explained only 16% of the total variance, indicating limited separation between rejection and non-rejection samples due to high background noise in global gene expression. After selecting the top 20 differentially expressed genes, PC1 accounted for approximately 80% of the variance, revealing that these genes capture the major transcriptional differences associated with graft rejection.

- *Heat-map*

The heatmap of the top differentially expressed genes reveals distinct transcriptional profiles between rejection and non-rejection liver transplant biopsies. Rejection samples show coordinated upregulation of immune-related genes, indicating a strong immune activation phenotype.

Overall, this expression pattern reflects a T cell– and NK cell–driven rejection phenotype, consistent with known mechanisms of immune-mediated graft injury. The clustering and functional enrichment reinforce that the selected genes capture the dominant biological processes underlying rejection in this dataset.

Modeling:

After cleaning our dataset, we needed to reduce the number of features as it currently far exceeded our number of samples. We used t-test, PCA, and Independent Component Analysis (ICA) to identify the genes and/or components that were either statistically important or were able to capture differences between our target groups. We tested the classifiers Logistic Regression, Random Forest, and XGBoost, and we found the combination that provided the best results was Logistic Regression combined with a reduced dataset of the top 500 genes contributing to one component of ICA. This model had good overall accuracy, a high AUC score, and high specificity, but had a significant false negative rate that could be improved upon.

Conclusions:

- Altogether, genes selected after data analysis (selected features) reinforce that our model is capturing a biologically meaningful signal — the immune activation underlying rejection.
- Logistic Regression + ICA Reduced (500 genes) model had the highest accuracy and ROC-AUC scores.
- Further optimization on the model with addition of more samples can improve the performance, and lead to less false negatives.

Limitations and Future Work:

- Due to the large number of features in our dataset, feature selection is needed to reduce the complexity. However, in doing so, this removed background signals or other noise that may have been useful in further categorizing the rejection class as proper rejection. By spending more time on feature selection and carefully fine tuning the target genes, we can ideally lower the false negative rate.
- After a liver transplant, patients are required to take multiple immunosuppressive medications to minimize the risk of graft rejection. These drugs are critical for graft survival, but the specific regimens can vary across transplant centres. The dataset used in this study includes gene expression profiles from patients across ten different centres; however, detailed information about their immunosuppressant regimens was not available. Variations in these treatment protocols may influence gene expression patterns. Additionally, information regarding patients' adherence to their prescribed medications was not accessible, representing another potential confounding factor beyond our control.

References:

1. Madill-Thomsen, K. S., Gauthier, P. T., Abouljoud, M., Bhati, C., Bruno, D., Ciszek, M., ... & Halloran, P. F. (2025). Defining an NK Cell–enriched Rejection-like Phenotype in Liver Transplant Biopsies From the INTERLIVER Study. *Transplantation*, 10-1097.